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Applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in Higher-Education Curriculum from the Perspective of Faculty Members: A Phenomenological Study

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Abstract

The current research was conducted with the aim of applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education curricula from the perspective of the faculty members. This research is guided by a qualitative approach and phenomenological design and by using a purposeful sampling method from a criterion sampling plan. The field of research included experienced faculty members of Iranian universities. The data collection tool was a semi-structured interview. Claysey's seven-step strategy was used for data analysis, and the interviews continued until the saturation of the findings. Validity review using Creswell and Miller criteria, peer review, and members review methods were used. The findings related to the executive challenges of implementing approaches based on ChatGPT in higher education curricula from the perspective of the faculty members indicated eight main-categories of "content obstacles", "cognitive obstacles", "psychological obstacles", "technical obstacles", "human obstacles", "communicational obstacles", "structural obstacles", "strategic obstacles", and the findings of the promotion and productivity factors of implementing procedures based on the applicability of ChatGPT in higher education curriculum from the perspective of the faculty members indicated nine main-categories of "psychological factors", "cognitive factors", "communicational factors", "content factors", "organizational factors", "human factors", "technical factors", "pedagogical factors" and "instructional designing factors". The author presented an agenda for educational research and practice in terms of the applicability of ChatGPT in higher education.

Keywords: Chat Generative Pre-Trained Transformer (ChatGPT), Higher education curriculum, Faculty members, qualitative approach, and phenomenological design

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Introduction

Artificial intelligence (AI), over the past decades, has practiced rapid development and has converted many territories of the community (Mhlanga, 2023; Albahri et al., 2023; Singh et al., 2024). Artificial intelligence has disposed itself at the forefront of investigation areas ranging from robotics and medicine to training and independent driving (Ahmed et al., 2023). Extensively, AI adverts to the capability of a computer system to imitate the manner of the human brain (Zhao et al., 2023). This encompasses processing information in the shape of external data, learning by teaching, and according to that learning, attaining the purposes for which it was deliberated (Brazdil & Jorge, 2001). The execution of AI in education has prepared an essential effect, substantiated by advancements in the proficiency of the educational cycle (Chiu et al., 2023; Alasadi & Baiz, 2023; Kizilcec, 2024), learning personalization (Dogan et al., 2023), the elevation of global learning (Jara & Ochoa, 2020), the improvement of educational management (Escotet, 2023), and the production of more intelligent content in consequences of proficiency and effectiveness (Lv, 2023; Wang et al., 2023). Nowadays, modern procedures of AI technology play a considerable role in the enhancement of the training–learning sequence. AI-aided education is not restricted to being a mere vintage but is a cycle in which learning moves beyond the ordinary attainment of knowledge. In this regard, artificial intelligence (AI) appears as a modern technology with major potential in the area of education, as artificial intelligence-oriented structures could nurture personalized learning and reconcile to the interests and demands of each trainee (García-Peña et al., 2020).

Artificial intelligence (AI) emergence has revolutionized disparate territories, comprising education and healthcare (Zahlan et al., 2023). Among the newest productions, the presentation of OpenAI from the Chat-Generative Pre-trained Transformer (ChatGPT) in November 2022 notifies a remarkable stride in the evolution of artificial intelligence technology (Dhankhar, 2024). This paradigm has harnessed the strength of artificial intelligence and natural language processing to imitate human-like conversation (Nazir & Wang, 2023; Hassani & Silva, 2023), providing both transformative challenges and potential across scientific territories, exhaustively in healthcare (Huang et al., 2024; Casheekar et al., 2024).

The application of ChatGPT in instruction has created remarkable interest due to its capacity to reinforce the learning experiences of trainees (Lai et al., 2023; Song & Song, 2023; Hasanein & Sobaih, 2023; Zirar, 2023). By supplying personalized and rapid responses, this structure has the capability to consider individual learner necessities, provide immediate feedback, and simplify the intricate notions of understanding (Albdrani & Al-Shargabi, 2023; Park, 2023; Ngo, 2023). In this sense, it regards a promising instrument that enhances the trainee's active cooperation and cognitive growth by adjusting to their learning style and suggesting continuous scaffolding in their knowledge acquisition cycle (Sánchez, 2023).

Following this pace, it has been ascertained to be noteworthy equipment to optimize the expansion of writing capabilities in trainees (Črček & Patekar, 2023; Esmaeil et al., 2023). By collaborating with this structure, learners can embrace grammatical modifications (Shaikh et al., 2023), and presentations for detailed and advanced feedback on their writing (Zou & Huang, 2023), which offers them the chance to promote their written communication and attain greater efficiency in their written situation (Farrokhnia et al., 2023). Accordingly, this application prepares itself as a precious assistance instrument for scientific writing (Rezaei et al., 2024; Rojas, 2024); nevertheless, it is necessary to comprehend that it should not be regarded as a perfect solution for the production of scientific content (Albdrani & Al-Shargabi, 2023; Bin-Nashwan et al., 2023; Xiao & Zhi, 2023). Scholars must practice their

experience and knowledge to accredit and accomplish the information prepared by the instrument (Hasanein & Sobaih, 2023). Meanwhile, it also surpasses its capability to simplify group debates and optimize trainee cooperative collaboration in assignments and schemes (Gill, 2024). This develops a sense of Sociability among trainees by permitting participation and an interchange of opinions (Nikghalb & Cheng, 2024; Chukwuere, 2024).

Several investigations have attempted to detect the potential effect of AI chatbot assistant models, particularly with the outstanding ChatGPT equipment for trainees in teaching (Kavadella et al., 2024; Ayoubi, 2024; Sun et al., 2024; Chan & Zhou, 2023; Mahmud et al., 2024). Diverse applications of ChatGPT in education were investigated, such as adaptive testing (Parker et al., 2024), personalized learning experiences (AlShaikh et al., 2024; Kiryakova & Angelova, 2023; Chan & Hu, 2023), and foreseeable analytics (Wang, Lund, et al., 2023). As a result, it demonstrates an inconceivable capacity to promote learning effectiveness and prepare personalized educational scaffolds for both trainees and instructors (Liao et al., 2024; Rasul et al., 2023). Likewise, it is vital to contemplate the hazards and restrictions accompanying these technologies, such as cultural diversities, data privacy, language domination, and accompanying ethical implications (Wang, Lund, et al., 2023). While ChatGPT has illustrated promising sufficiencies in miscellaneous medical applications, comprising some positive consequences in medical licensing excavations, recent surveys, such as the one carried out on the pharmacist licensing verification in Taiwan, highlight the necessity for accurate observation of its constraints in distinguished evaluation platforms (Wang, Shen, et al., 2023; Zong et al., 2023).

Purpose of the study

Many researchers have reviewed the potential advantages and concerns for the incorporation of ChatGPT in training. Nevertheless, an exhaustive overview is still lacking that can institute a theoretical foundation for empirical surveys aiming to harness the potential of this modern artificial intelligence technology. Such an overview can present a detailed analysis of ChatGPT's efficiencies and obstacles according to scientific literature and depict the plausible occasions and threats to instruction.

Research Questions

- RQ1: What are the challenges and obstacles of implementing approaches based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education curricula from the perspective of the faculty members?
- RQ2: What are the factors in the field of promotion and productivity of the implementation procedures based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education curriculum from the perspective of the faculty members?

Method

The present deductive research was conducted to represent the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education curricula from the perspective of faculty members by implementing a qualitative research design and using the phenomenological method. According to Akbari & Ghafourian (2014), this key assumption is that if phenomena are examined in the context of occurrence, with the eyes of the individuals who experience it, many of their hidden angles will emerge and ultimately lead to deep knowledge, and the other hand, tools for future research and operationalization of conceptualized symbols are provided. In this method, "the researcher tries to understand and explain the meanings of a concept or event from the perspective of some participants'

opinions about their lived experiences". Therefore, this investigation is trying to provide an in-depth description of the challenges and factors of the field of promotion and productivity based on the lived experiences of faculty members.

Participants

In this research, the participants included all academic faculty members in the universities of Tehran state located in Iran, who were working in the academic year of 2023-2024. Also, a purposeful sampling plan and a criteria sampling method were used. After the interview with the twelfth participant, the theoretical saturation of the data was confirmed, but to validate the findings, the interview process continued until the fifteenth subject. The selection of participants in the process of data collection in this research included: 1- Willingness and appetite to participate in the interview process 2- Attaining the required level of experience for this research 3- Being relevant to the educational level based on the educational activity 4- Possessing the necessary subject expertise. Table 1 shows the demographic characteristics of the research field participants.

Table 1 Demographic characteristics of the research field participants

Code	Gender	Teaching experience	Educational major	Position	Educational Degree
A1	Female	19	Psychology	Assistant Professor	P.H.D
A2	Male	5	Biology	Assistant Professor	P.H.D
A3	Male	4	Educational Sciences	Assistant Professor	P.H.D
A4	Female	16	Educational Management	Associate Professor	P.H.D
A5	Male	29	Educational Sciences	Assistant Professor	P.H.D
A6	Female	12	Psychology	Assistant Professor	P.H.D
A7	Male	11	Educational technology	Associate Professor	P.H.D
A8	Male	14	Psychology	Associate Professor	P.H.D
A9	Female	31	Educational Management	Professor	P.H.D
A10	Female	23	Curriculum planning	Assistant Professor	P.H.D
A11	Male	16	Educational Sciences	Associate Professor	P.H.D
A12	Female	21	Biology	Assistant Professor	P.H.D
A13	Male	26	Educational Sciences	Associate Professor	P.H.D
A14	Female	18	Educational technology	Associate Professor	P.H.D
A15	Male	11	Computer	Assistant Professor	P.H.D

Data collection and analysis

The data collection tool was a semi-structured interview. For data analysis, Claysey's (1997; Ghasemi and Salehi, 2016; Azimi et al., 2024; Zabet et al., 2023) seven-step strategy was used according to the following details.

A) All the descriptions provided by the participants in the study, which is commonly called the protocol, were studied to supply double familiarity; B) With the aim of extracting key categories, referred to each protocol, and concepts and categories that were directly related to the event under study were extracted; C) In the next step, with the aim of formulating the meanings of the underlying sentences, it was tried to understand the meaning of each of the main categories; D) In a continuous and rotating process, all the above steps were repeated for all 15 protocols, and an attempt was made to classify and organize the formulated and aligned meanings in clusters of main categories (within the framework of

challenges and factors of the field of promotion and productivity of the implementation procedures based on applicability of ChatGPT in higher education); E) An effort was made to combine the results in the framework of a comprehensive and exhaustive description of the subject under investigation; F) With the aim of making a clear and explicit statement of the underlying structure of the phenomenon under study (the challenges and factors of the field of promotion and productivity of the implementation procedures based on applicability of ChatGPT in higher education), an inclusive description of the phenomenon under study was carried out; G) Finally, for the final validation of the obtained findings, it was tried to make the necessary corrections and adjustments by repeatedly referring the findings to the interviewees and to finalize the findings that can be presented. In this regard, by using the narrative data obtained from the interviewees, while gaining a deeper understanding of the event under study from their viewpoint, it was tried to make the results meaningful and increase the credibility of the data.

Validity and Credibility

In order to ensure the validity of the findings made from the qualitative analysis, based on the criteria provided by Creswell & Miller (2000), from the method of peer review (two professors who were experts in the field and knowledgeable about the research issue in the territory of data coding and interviews) and the member's revision method (three interviewees) was used.

Findings

The themes obtained from the data of this research are all about the experiences of the faculty members as index agents of the higher education system from the phenomenon of ChatGPT in the higher education curriculum, and based on the objectives of the study. In general, these findings were arranged in two sections of the challenges and obstacles of the implementation procedures based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education from the perspective of the faculty members with 50 basic themes and 8 main-organizing theme as index organizing categories and also in the section of the factors in the field of promotion and productivity of the implementation procedures based on applicability of ChatGPT in higher education with 62 basic themes and 9 main-organizing categories. In the following, each of the findings is explained and analyzed under the relevant questions.

- **RQ1: What are the challenges and obstacles of implementing approaches based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education from the perspective of the faculty members?**

The findings of the study and examination of the challenges and obstacles of implementing approaches based on the Chat Generative Pre-Trained Transformer (ChatGPT) applicability in higher education from the perspective of the faculty members are presented in Table 2.

Table 2 The challenges and obstacles of implementing approaches based on the applicability of ChatGPT in higher education

Overarching theme	Main-organizing themes	Basic themes	Code
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The challenges and obstacles of implementing approaches based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education	Content barriers	Providing poor-quality educational content	A2, A4
		Absence of codified AI technological curriculum in higher education	A2, A10, A8
		Lack of content information in the field of AI virtual education	A4
		Incompatibility of existing curricula structure and AI technological procedures	A12, A11
		Lack of supportive AI technological content production packages	A13
		Lack of reliable generative AI content databases	A2
		Failure to provide attractive content	A4, A11
		Lack of constructive generative AI curriculum	A8
		Lack of detailed analysis of AI-based information and communication technology content space for instructors	A13
		Lack of detailed analysis of emotional literacy components in existing generative AI curriculum activities	A14
	Cognitive barriers	Paying too much attention to the goals of the cognitive domain	A7
		Lack of Staff AI technological literacy	A5,A7,A10
		Lack of common AI media and digital literacy of learners	A8,A2,A4,A13,A10
		Failure of educational system officials to follow up on instructors' AI technological literacy	A15
		Lack of sufficient understanding of the execution procedures of ChatGPT platforms in education	A10,A7
	Psychological barriers	Lack of definition of applied AI technological literacy in instructors' work curriculum	A7
		Lack of valuing and internalization of values in instructors' behavior	A5
		Instructors' educational motivational weakness	A6
		Lack of emotional-perceptual skills in generative AI technology contexts	A3, A13
		Lack of mighty motivation of learners in cooperation and participation	A13
		Lack of Learners' interest in existing boring procedures	A9, A15
		Inability to identify students' inner feelings in the context of AI technology	A14
	Technical barriers	Failure to provide an intense Internet platform	A2, A15, A4, A11, A13
		Absence of a dedicated official education platform based on the generative AI	A4, A5, A9, A13, A11, A14
		Lack of valid time facilities	A4
		Weakness of educational generative AI technology equipment	A9, A6, A5

	Human barriers	Limitation of relevant facilities in the field of universities' educational AI technologies	A15, A14
		Lack of the generative AI customized systems for students	A2, A4
		Weakness of educational systems in accessibility	A9
		Lack of AI advanced infrastructures	A8, A5
		Failure of educational institutions to provide generative AI technology executive facilities	A7
		Lack of access to free educational platforms	A12
		The absence of cooperation with efficient specialists	A1, A2
		Confusion among supervisors, instructors, and teaching staff	A4
		Incapability to identify the basic needs of learners in generative AI web platforms	A10
		The inability of instructors to fully manage the internet space	A5, A14
		The inability of instructors to control the AI technological environments	A10
		Being unfamiliar with existing AI technological tools	A13
		Lack of instructors' awareness of ChatGPT literacy components in the context of technology	A14
		Lack of mastery of teaching staff from AI-based active teaching-learning approaches	A12
	Communication barriers	Being incapable of making Communication skills	A1, A3, A10
		Restrictions on the use of AI communication platforms	A8
		Reducing AI communication space between instructor and learners	A8
	Structural barriers	Weakness of AI financial and Affordable management	A9, A6, A14
		Weakness of AI managerial and planning infrastructures	A6
		The existence of hierarchical and cultural differences among students in the terms of generative AI learning	A1
		Weakness of existing cultural procedures from the use of AI educational technologies	A14
	Strategic obstacles	Lack of necessary practical AI training courses	A1
		Failure to provide factual practical skills in AI educational package production platforms	A1
		Making contextual conflicting decisions	A7

The findings obtained from the study and examination of challenges and obstacles of implementing approaches based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education in the form of 8 main-categories ("Content

barriers", "Cognitive barriers", "Psychological barriers", "Technical barriers", "Human barriers", "Communicational barriers", "Structural barriers", "Strategic barriers") and 51 sub-categories were presented in Table 2. Chart 1 clarified index factors of challenges and obstacles of implementing approaches based on the ChatGPT applicability in higher education.

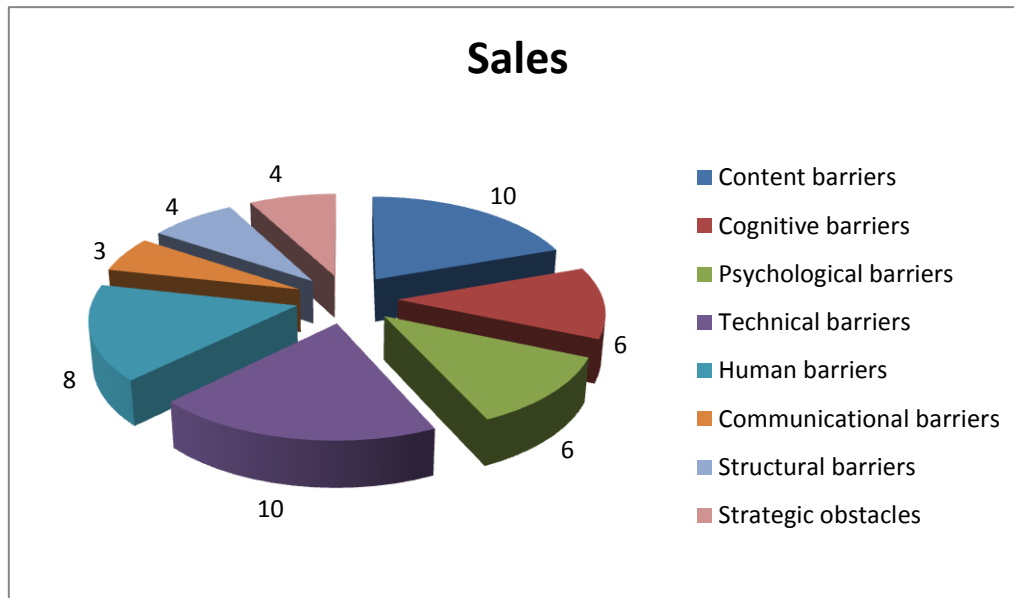


Chart 1 Index factors

- **RQ2: What are the factors in the field of promotion and productivity of the implementation procedures based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education from the perspective of the faculty members?**

The findings of the study and examination of factors in the field of promotion and efficiency of the implementation procedures based on the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education from the perspective of the faculty members were presented in Table 3. In the following, each of the identified topics and factors were described.

Table 3 Factors in the field of promotion and productivity of the implementation procedures based on the ChatGPT applicability in higher education

Overarching theme	Organizing theme	Basic themes	Code

The factors in the field of promotion and productivity of the implementation procedures based on the ChatGPT applicability in higher education	psychological factors	Providing solutions to strengthen the self-esteem of learners in utilizing ChatGPT	A6
		Taking advantage of incentive approaches	A2, A10
		Paying attention to the emotional needs of students	A12, A15
		Trying to institutionalize generative AI values	A10
		Commitment and responsibility in learners	A5
		Identifying the components of emotional literacy in the generative AI web context	A10
		Setting ChatGPT educational programs according to emotional literacy	A12
		Promoting motivational generative AI content production approaches	A15
		Using AI hypothetical situations to encourage the exchange of feelings between individuals	A6
		Creating a healthy moral identity of the ChatGPT applicability	A3
		Following the motivational strategies for learners	A1, A8
		Providing creativity supplies	A5
		Management of the emotional components and how to deal with it	A13, A15
	Cognitive factors	Promotion of ChatGPT technological professional knowledge (TK)	A12
		Improving ChatGPT technological content literacy (TCK)	A2, A5, A7
		Providing the necessary generative AI supportive cognitive packages	A12
	Communicational factors	Creating a continuous communicational environment	A3
		Strengthening ChatGPT virtual communicational networks	A7
		Providing contexts for positive social interaction	A6
		Establishing ChatGPT managed communicational procedures	A3
		Providing learning interaction situations	A6
	Content factors	Creating solutions to increase the quality of ChatGPT content production packages	A10
		Application of ChatGPT generative technology-based educational content	A2
		Increasing the literacy of AI standard content production	A13, A1

		Considering ChatGPT's hidden curriculum features	A15
		Providing AI attractive content production packages	A5
		Strengthening the procedures for providing ChatGPT content production	A7
		Creating AI learner-oriented content production packages	A15
	Organizational factors	Equipping dynamic generative AI digital cultures	A4, A14
		Creating generative AI social practical platforms	A15
		Modeling effective AI social behavior	A6
		Providing suitable policies through generative AI education	A14
		Training human resources with AI technological knowledge (TK)	A4, A14
	Human Factors	Adaptation and adjustment of instructors in AI educational procedures	A15
		Recruitment of human resources with generative AI technological literacy	A1
		Providing the position of AI teaching assistant as a strategy for the learning process	A5
		Increasing complete mastery of the generative AI-based web environment by human resources	A5
		Creating procedures for managing the presence of learners	A11
	Technical factors	Strengthening the audio and visual capabilities of generative AI technology platforms	A3
		Increasing the flexibility of using generative AI multimedia systems	A3
		Removal of AI generative systems providing fees for users	A3
		Providing generative AI flexible platforms that strengthen the acceptance of ChatGPT applications by indicating TAM theory	A1
		Equipping educational centers with AI generative platforms	A8, A14
		Providing compiled AI training courses that can be customized with the web platforms	A1
		The presence of educational facilities based on active generative AI	A4
		Providing special AI generative facilities for higher education offices	A3, A8
	pedagogical factors	Promoting critical practices	A15
		Increasing study space procedures of generative AI technology tools	A2
		Revision of existing teaching techniques and changes in the role of instructor and student in the AI technology context	A13
		Using student-centered generative AI adaptive	A8, A13

	Educational design factors	learning styles	
		Increasing in-service AI courses for trainers	A9, A11, A12, A14
		Teaching the language of generative AI digital space	A1
		Teaching substantial learning of generative AI tools	A5
		Application of innovative ChatGPT technological teaching plans	A2
		Arranging necessary ChatGPT training programs	A9
		Appropriateness to the level of goals and educational needs	A9
		Increasing and strengthening ChatGPT executive procedures	A9
		Providing a ChatGPT platform for knowledge transfer	A9
		Providing diverse AI learning styles	A4
		Estimating the necessary standards of the AI educational content packages	A5
		Creating harmony between the two dimensions of generative AI content education and the learning experiences	A11

The findings from the study and investigation of the Factors in the field of promotion and productivity of the implementation procedures based on the ChatGPT applicability in higher education from the perspective of the faculty members in the form of 9 main-organizing categories "psychological factors", "cognitive factors", "communicational factors", "content factors", "organizational factors", "human factors", "technical factors", "pedagogical factors", and "educational design factors" and 61 basic themes were presented in Table 3. Chart 2 clarified index factors in the field of promotion and productivity of the implementation procedures based on the ChatGPT applicability.

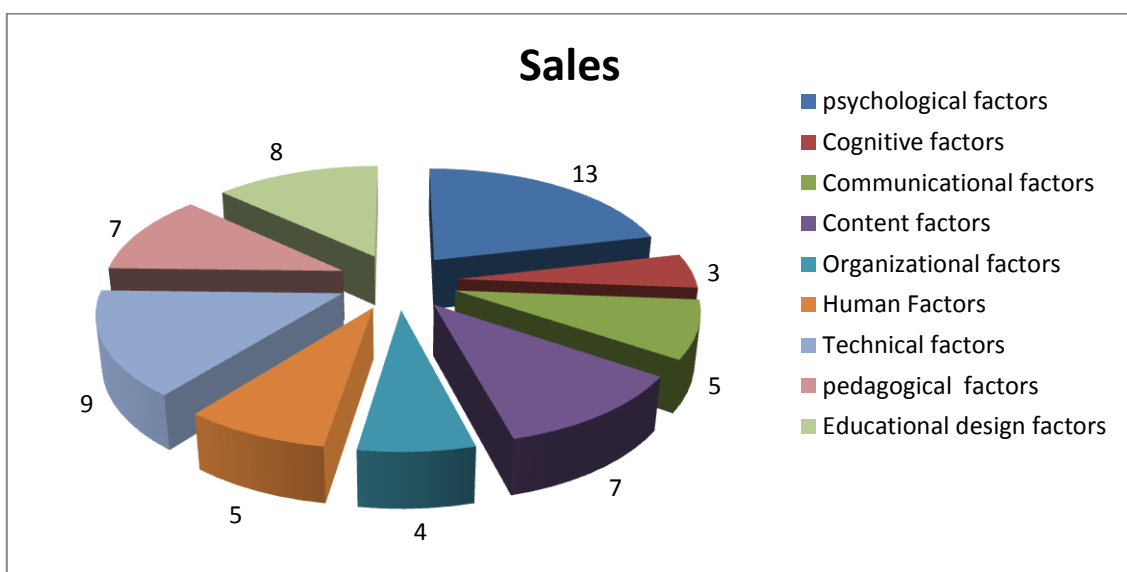


Chart 1 Index factors

Discussion, conclusion, implications, and recommendations for future practice

The phenomenological analysis of ChatGPT illustrated that this modern AI technology has miscellaneous potential utilizations for education, but also comes with determined challenges, as presented in Figure 1. The outcome of the phenomenological analysis could assist in considering this substantial question: "What is the applicability of Chat Generative Pre-Trained Transformer (ChatGPT) in higher education from the perspective of the faculty members?" In the following, we analyzed a couple of scenarios that can be regarded as also suggested by Farrokhnia et al. (2023).

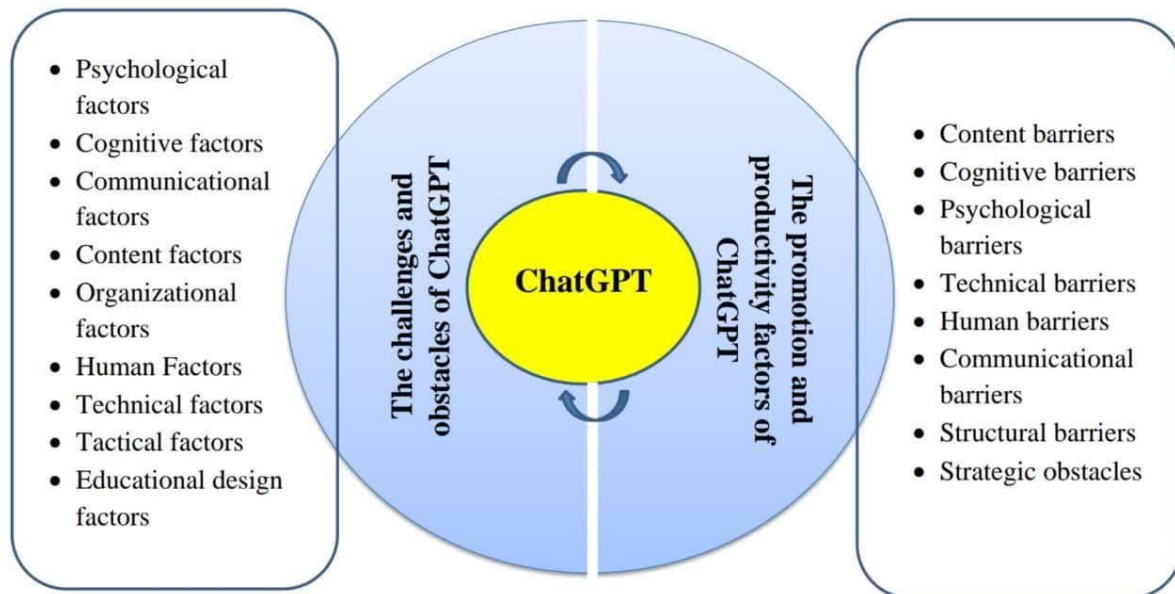


Fig 1 Conceptual framework of ChatGPT applicability in higher education from the perspective of faculty members

In this section, we analyze the themes of both research scenarios from the perspective of examining the main-organizing themes that have the highest frequency of repetition.

To answer the first question of the research, the results emphasized the 50 primary themes that were organized in a process of conceptual construction and payment in eight main axes including: "content barriers", "psychological barriers", "cognitive barriers", "technical barriers", "communicational barriers", "human barriers", "strategic barriers", "structural barriers". The results supplied from the investigation of the challenges and obstacles indicators illuminated the existence of the most considerable stage of indices "content barriers", and "technical barriers". In this regard, to analyze the theme of "technical barriers", Gao et al. (2023) illuminated that ChatGPT lacks a profound knowledge of the meaning of the words it reflects (J. Gao et al., 2023). It distinguishes models and produces plausible reactions, but it does not totally grasp the notions behind the words (Bogost, 2022). This may result in responses that are sometimes lacking depth and insight (Borji, 2023) and potentially off-topic (Gupta et al., 2023), particularly for performing tasks that require a nuanced understanding of specific domain knowledge (Dimitrov, 2023).

Regarding the "content barriers", according to the researcher, one of the most important underlying obstacles and challenges was the category of "the absence of codified AI technological curriculum in higher education" with the maximum code obtained in this determinant, because as one of the key components of the foundation, it has the greatest

possible impact among categories of this classification of challenges, and the other results of this category respectively included "Lack of detailed analysis of emotional literacy components in existing AI-curriculum activities", "Providing poor-quality educational content", "Lack of content information in the field of AI virtual education", "Incompatibility of existing curricula structure and AI technological procedures", etc.

The results showed that, in the investigation of "cognitive barriers", from the researcher's viewpoint, the main identified challenges are related to the affairs that emphasize the generative AI curriculum. This finding can be attributed to the categories of "Lack of staff AI technological literacy", "Lack of common AI media and digital literacy of learners", "Failure of educational system officials to follow up on instructors' AI technological literacy", etc; In the analysis of "psychological barriers", it can be said that this category mainly describes the dimensions of various obstacles and challenges of the psychological factors of ChatGPT applicability in higher education from the perspective of higher education faculty members and examined the field of technology in terms of dimensions of psychological literacy. The researcher identified the two categories "Lack of emotional-perceptual skills in generative AI technology platforms" and "Instructors' educational motivational weakness" in AI platforms, which is a psychological result of not knowing enough about these learning environments, and the other results of this category respectively included "Inability to identify students' inner feelings in the context of AI technology", "Lack of Learners' interest in existing boring procedures", and "Lack of strong motivation of learners in cooperation and participation"; The "structural barriers" describing the challenges of ChatGPT applicability in the field of implementation hierarchy included "Weakness of AI managerial and planning infrastructures" and the affairs that affect this category including "Weakness of AI financial and affordable management"; The field of human obstacles emphasizes the issues related to organization plans of human resources in the field of higher education, which mainly included evidence such as "Absence of cooperation with efficient specialists", "Confusion among supervisors, instructors and teaching staff", "the inability of instructors to control the AI technological environments". Also, the sub-categories that describe the challenges in the field of human resources included themes "Lack of mastery of teaching staff from AI-based active teaching-learning approaches", "Being unfamiliar with existing AI technological tools", "Inability of instructors to fully manage the internet space", and "Incapability to identify the basic needs of learners in generative AI web platforms", etc; In the field of "strategic obstacles", these cases mainly categorized strategies that challenged the level of ChatGPT applicability of implementation procedures, which included the themes of "Lack of necessary practical AI training courses" and "Failure to provide factual practical skills in AI educational package production platforms"; In considering of communicational barriers, it can be contemplated that the category of "Being incapable in making Communicational skills" indicated that the current human resources were not equipped to use Communicational skills under AI technology. Also, the two categories "Restrictions on the use of AI communication platforms" and "Reducing AI communication space between instructor and learners" indicated the lack of available platforms and tools based on the technology network in providing dynamic communicational factors and indicators.

findings from the second scenario of the research illuminated that the main themes extracted based on the participants' viewpoints, contained nine main-categories "psychological factors", "cognitive factors", "communicational factors", "content factors", "organizational factors", "human factors", "technical factors", "pedagogical factors" and "educational design factors". In this sense, the results obtained from the investigation of the factors in the field of promotion and productivity of the implementation procedures according to the ChatGPT applicability in higher education demonstrated the existence of considerable stage of index of "psychological factors" following "technical factors"; the determinant of

psychological factors dedicates to those factors that consider academic self-esteem (Bin-Nashwan et al., 2023); incentive approaches (Mungoli, 2023); the emotional needs of students (Yu, 2024); to institutionalize AI values (Kshetri, 2024); Commitment and responsibility in learners (Mhlanga, 2023; Huallpa, 2023; Lindebaum & Fleming, 2024); motivational AI content production approaches (Kusal et al., 2021); AI hypothetical situations to encourage the exchange of feelings between individuals (Ho et al., 2023), and other identified themes also included "Identifying the components of emotional literacy on the AI web context", "Setting ChatGPT educational programs according to emotional literacy" and etc; the field of technical factors provides all the space required for the effective and efficient presence of learners in order to recognize the components of ChatGPT applicability. Also, this category of factors identify themes that cause the removal of restriction options, such as "Removal of generative AI systems providing fees for users"; the field of educational design factors was recognized as one of the cases that can play a significant role in the design of special operational lesson plans at the macro and micro level for the instructor as the executive of educational design at the classroom level; the field of content factors specifies the items that describe the fields of design, implementation and evaluation of content indicators. The researcher stated that one of the most substantial themes of this classification of topics is "Increasing the literacy of AI standard content production" counted as one of the basic and vital themes of this spectrum, and the other results of this category respectively included "Creating AI learner-oriented content production packages" (Habibi et al., 2023; Albdarani & Al-Shargabi, 2023), "Creating solutions to increase the quality of ChatGPT content production packages" (Castillo-González et al., 2022; Liu et al., 2023), "Providing attractive AI content production packages", and "Strengthening the procedures for providing ChatGPT-content production"; pedagogical factors are equipped with mechanized tactician training brigades and provide decentralized self-propelled remedial training system and prepare the platform for efficient and effective systematic activities; In describing the conditions of human factors, the category of "Recruitment of human resources with generative AI technological literacy" (Smythe et al., 2021; Deepa et al., 2024) is known as the indicator theme of this spectrum, and the other results of this category respectively included "Increasing complete mastery of the generative AI-based web environment by human resources" (Fütterer et al., 2023), "Training human resources with AI technological knowledge" (Chiu, 2024) as the outputs of this process which are capable to have "Adaptation and adjustment of instructors in AI educational procedures"; The main theme of communicational factors refers to appropriate and efficient interaction of characteristics, and one of the most important themes in this dimension is "Establishing ChatGPT managed communicational procedures" (Šedlbauer et al., 2024) in the context of technology following by themes of "Strengthening ChatGPT virtual communicational networks" (Fütterer et al., 2023), "Providing learning interaction situations" (Tossell et al., 2024), and "Providing contexts for positive social interaction" (Deepa et al., 2024; Belghith et al., 2024; Abdullah et al., 2022); moreover, the main category of organizational factors refers to the macro and systemic dimensions of ChatGPT applicability in higher education from the perspective of faculty members. The objective characteristic of this explanation is emphasized by the theme "Providing suitable policies through generative AI education" (Chan & Hu, 2023; Hasanein & Sobaih, 2023). This factor will be a manifestation of the technical and executive structure of the pre-trained personalized artificial intelligence curriculum because the decision-making results in such issues are reflected in the existing educational curricula. The output of this series of cases will be factors such as "Creating generative AI social practical platforms" (Chan & Hu, 2023; Chiu, 2024), "Modeling effective AI social behavior" (Singer et al., 2023; Choi et al., 2024; Al-Obaydi et al., 2023), and "Equipping dynamic generative AI digital cultures"; ultimately, in the analysis of cognitive factors, one of the components that play a

significant role in increasing the scope of ChatGPT applicability in higher education in the AI-based technological curriculum is "Improving ChatGPT technological content literacy" (Chiu, 2024; Jing et al., 2024; Kelly et al., 2023). With sufficient knowledge of human resources from these platforms, mastery, and dominance of how to use these tools and devices available in the ChatGPT technological platform will be provided, and the teacher will be able to control the existing space and provide solutions based on ChatGPT applicability. Paying attention to these factors will fulfill the "Promotion of ChatGPT technological professional knowledge" (Jing et al., 2024; Kohnke et al., 2023). Also, the upstream educational system will be to "Provide the necessary generative AI supportive cognitive packages" (Kiryakova & Angelova, 2023; Habib et al., 2024) to create the ground for the improvement and flourishing of instructor's ChatGPT applicability literacy.

In the end, it should be indicated that while phenomenology analysis can present an in-depth analysis of the ChatGPT applicability in higher education, it has restrictions in prioritizing the affairs recognized in each classification, as also indicated by Leiber et al. (2018). Accordingly, empirical investigations with quantitative procedures including the best-worst strategy (see Farrokhnia, 2015) are needed to expand the recent survey findings. This can be attained through in-depth interviews with pertinent experts to specify the weights and significance of the obtained weaknesses and opportunities.

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